Lecture 1

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$$R = \frac{dX}{dt}$$

$$r = \frac{d\xi}{dt}$$

$$\xi = \frac{X}{V}$$

$$r \times V = \frac{dX}{dt}$$

$$\frac{N_j - N_{j0}}{\alpha_j} = X$$

$$\downarrow$$

$$\frac{dN_j}{dt} = \alpha_j \frac{dX}{dt}$$

$$= \alpha_j V r$$

$$N_j = c_j V$$

$$\frac{d(c_j V)}{dt} = \alpha_j r V$$

$$\downarrow$$

chain rule

$$\downarrow$$

$$\frac{dc_j}{dt} = \alpha_j r - c_j \frac{d \ln V}{dt}$$

$$\left\{ \frac{1}{V}\frac{dV}{dt} = \frac{d}{dt} \ln V \right\}$$

$$\frac{dn_j}{dt} = \alpha_j r - c_j \frac{d \ln V}{dt}$$

in terms of $\varphi_k \quad \left(\varphi_k = \frac{x}{V} \right)$

$$\frac{dx}{dt} = \frac{d}{dt}(\varphi_k V) = rV$$

$$\therefore \quad \frac{d\varphi_k}{dt} = -\varphi_k \frac{d \ln V}{dt} + r$$